Dr Margaret Oakley Dayhoff: The mother and father of bioinformatics

In the 21st century one must have heard about the term “bioinformatics”. Especially for people across sciences, working with large biological datasets, these tools prove to be an ultimate boon, helping to analyse and find significant results just sitting at one’s computer. American biophysicist Dr Margaret Oakley Dayhoff was one of the founding members of this discipline, providing it with its required direction, back in the 1960s. A period when computers were not so developed and women researchers were highly marginalized. Still, she was able to practise science, make pioneering contributions, and fulfil her duties as a mother, at the same time being treated as an outsider.

In her early life, signs of her academic promise were quite evident. She was valedictorian (class of 1942) at Bayside High School, Bayside, New York, and from there received a scholarship to Washington Square College of New York University, graduating magna cum laude (with great distinction) in mathematics in 1945. That same year, Dayhoff entered Columbia University to get her PhD in quantum chemistry under the mentorship of prominent chemist and World War II operations researcher George Kimball. Her acceptance was a rarity for the time. After WWII, more men entered the sciences, and chemistry became even more male-dominated than in the previous decade, with only five percent of chemistry PhDs going to women, down from eight percent. Her management of her research data was so impressive that she was awarded a Watson Computing Laboratory Fellowship. As a fellow at the Watson IBM Computing Laboratory in 1947–1948, she used punch card machines to calculate resonance energies in small molecules (Bruno J Strasser,2010). Her PhD thesis used punch-card computing to calculate the resonance energies of organic molecules. The research that she undertook as a graduate student was published, with Kimball as coauthor, in 1949 in the *Journal of Chemical Physics.* At this time, fewer than 10% of PhD candidates were women, and the numbers were even lower in the sciences. She navigated these initial stages of her scientific journey and training, while mostly being treated as an outcast most of the time. But all the required seeds for some great innovations were already sown.

In between all this in 1948, she got married to Edward Dayhoff, a student in experimental physics whom she had met at Columbia. According to Rachel Levy’s article on wordpress and her conversations with the Dayhoff family, Edward was endlessly supportive of her wife's ambitions. The pair had two daughters together. Margaret took a few years off from her career to raise her two daughters after completing postdoctoral fellowships at the Rockefeller Institute (1948 to 1951) and University of Maryland (1957–1959). Although she had clearly shown her skills in this evolving field, when she returned to research and began applying for grants to fund her work in 1962, she was met with a shock. The National Institute of Health (NIH)

rejected her application, with the explanation that “[Dayhoff] has been out of touch for some time … with this complicated and rapidly advancing area,” as historian Bruno Strasser wrote in his book *Collecting Experiments: Making Big Data Biology*. This kind of uphill climb for women who have taken time off to raise children was just one of the ways that scientific institutions hindered women’s advancement in those days.

Despite the NIH’s lack of support, Dayhoff was about to make her most important contributions during this upcoming decade. In 1960, she accepted an invitation from Robert Ledley, a pioneering biophysicist whom she met through her husband, to join him at the National Biomedical Research Foundation in Silver Spring, Maryland. Once in Maryland, Dayhoff had free will to use Georgetown University’s brand-new IBM 7090 mainframe with computing speeds six times faster than previous models. In 1965 she published the *Atlas of Protein Sequence and Structure*, a compilation of all known protein sequences, which has been cited over 4,000 times. She developed one-letter codes for amino acids that enabled computer sequencing and sorting, and an innovative database of nucleic acids to enable researchers to better understand protein structures. Her work has been foundational to the sequencing of many important proteins, including insulin. Later she used this knowledge to answer questions on evolution. She ran a computer analysis of proteins in a wide variety of species, from the candida fungus to the whale. Then she used their differences to determine their ancestral relationships. In 1966, with the help of Richard Eck, Dayhoff created the first reconstruction of a phylogenetic tree. In a 1969 *Scientific American* article, “Computer Analysis of Protein Evolution,” Dayhoff presented to the public one of these trees along with her research using computers for sequencing proteins. “Each protein sequence that is established, each evolutionary mechanism that is illuminated, each major innovation in phylogenetic history that is revealed will improve our understanding of the history of life,” she wrote. She was trying to show the life sciences community the potential of computerized models. “The culture of statistical analysis, let alone of digital computing, was completely foreign to most [biochemists],” explains Strasser in an interview with *Smithsonian.com*. “Some even prided themselves in not being ‘theorists,’ which is how they understood data analysis using mathematical models.”
With her experience and pioneering contributions, Dayhoff went on to occupy leadership positions in a scientific community historically dominated by men. She became the first woman to serve as an officer of the Biophysical Society, eventually serving as its secretary and president.

Unfortunately she died in 1983, at the age of 57 from a heart attack, while she was still actively pursuing science. Her legacy echoes in various forms. A fund was established after her death, in 1984, Margaret O. Dayhoff Award, to be awarded to exceptional women in biophysics. It is one of the top national honors in biophysics in the USA. Her seminal contributions as the mother of the science of bioinformatics, now routinely used as part of the process for naming bacteria, were acknowledged

with a bacterium being named after her in 2020, *Enemella dayhoffiae*. Many publications in experimental biology, which contains a combination of new experimental data and inferences drawn from comparisons with other data made available in a public database, uses an approach that Dayhoff had already started half a century ago.